ECON 381: Homework 3

Fall 2026

Due: Monday, October 19, 2026, at 11:59 p.m. Eastern Time
Submit electronically through the Canvas **Assignments** section.

Instructions and submission guidelines

1. Complete **all eight required questions**: two empirical textbook exercises (E4.2 and E5.1), two analytical textbook exercises (5.2 and 5.8), and four true-or-false questions. This homework covers Chapters 4 and 5 of the textbook (equivalently, Handouts 5 and 6).

2. You may submit individually or in a group of two, as allowed by the syllabus. A pair should make **one group submission**, list both names at the top of the homework PDF and in comments at the top of the do-file, and follow the submission instructions below. Group members receive the same grade.

3. Submit **all three** of the following on Canvas:

   - a **single PDF of handwritten solutions** for the analytical and true-or-false questions;
   - a **Stata do-file** containing your code for E4.2 and E5.1; and
   - a **Stata log file or PDF of printed Stata results** for the empirical exercises.

   For the empirical exercises, you may either **write your answers by hand** and include them in the homework PDF or **write your answers and interpretations as comments in the do-file**, as in Assignment 1. Label every exercise and subpart clearly. You must construct the scatterplot requested in E4.2, but you do not need to include it in your submission. You may include it in your homework PDF or Stata output PDF if you wish. **No credit** will be given for the empirical exercises if either the do-file or the log/results file is missing.

4. Written answers submitted in the homework PDF must be **handwritten**; they may be written on paper and scanned or written on a tablet and exported as a PDF. The submission must be **clear and legible**. Photos taken by phone or camera are often difficult to read and may be treated as illegible. **Illegible submissions will receive no credit**. Make sure that all pages and answers are clearly reproduced and included. The files you upload will be treated as your final submission; you are responsible for their completeness and legibility.

5. Use the same filename base for all submitted files:

   Individual: `ECON381_HW3_LastName`
   Pair: `ECON381_HW3_LastName1_LastName2`

   Append

   - `.pdf` for the handwritten solutions,
   - `.do` for the do-file,
   - either `.log` for the Stata log or `_Output.pdf` for printed Stata results.

Submission guidelines (continued)

6. Late work is accepted according to the syllabus policy. The percentage below is the share of the earned score retained:

Time submitted after the deadline	Percentage of earned score retained
At most 3 hours	80%
More than 3 and at most 6 hours	60%
More than 6 and at most 9 hours	40%
More than 9 and at most 12 hours	20%
More than 12 hours	0%

   Plan ahead when you anticipate a conflict; illness, university events, and other documented circumstances do not ordinarily excuse a late submission.

7. Any use of generative AI must follow the syllabus disclosure and attribution policy. If you use such a tool, include the required acknowledgment naming the tool and provider and explaining the purpose for which it was used.

Data and Stata instructions

- I will attach `Earnings_and_Height.zip` in the Canvas module **Textbook Dataset**. Unzip this file to get the dataset for both E4.2 and E5.1. Consult `Earnings_and_Height_Description.pdf` for the detailed data description.

- Use a do-file to organize all code. It must run from beginning to end without errors. Use comments to label exercises and subparts; if you answer empirical questions in the do-file, place each answer or interpretation in a comment immediately after the relevant command, as in Assignment 1.

- Record the commands and results in a Stata log, or print your Stata results as a PDF using `File` → `Print` → `Results`. On macOS, the `File` menu is in the menu bar at the upper left of the screen.

- Tutorials are on Canvas under **Stata Resources** → **Stata Demo Materials Repository**, including `STATA Basics.pdf`. To execute selected do-file lines, use `Shift + Command + D` on macOS or `Ctrl + D` on Windows. For Stata questions, contact the TA, Bowen Jiang, at bjiang12@vols.utk.edu.

1 Empirical Exercises

Source acknowledgment. Analytical Exercises 5.2 and 5.8 and Empirical Exercises E4.2 and E5.1 in this assignment are reproduced or adapted from James H. Stock and Mark W. Watson, *Introduction to Econometrics*, 4th edition (Pearson). Exercise 5.2(e) is omitted, Exercise 5.8 is modified for large-sample normal inference, and E5.1(d) is not assigned.

Empirical Exercise E4.2 (adapted from Stock and Watson, p. 129)

The data file `Earnings_and_Height` contains data on earnings, height, and other characteristics of a random sample of U.S. workers. A detailed description is given in `Earnings_and_Height_Description`. In this exercise, you will investigate the relationship between earnings and height.[1]

(a) What is the median value of height in the sample?

(b) (i) Estimate average earnings for workers whose height is at most 67 inches.

(ii) Estimate average earnings for workers whose height is greater than 67 inches.

(iii) On average, do taller workers earn more than shorter workers? How much more?

(c) Construct a scatterplot of annual earnings (*Earnings*) on height (*Height*). Notice that the points on the plot fall along horizontal lines. There are only 23 distinct values of *Earnings*. Why? *Hint: Carefully read the detailed data description.*

(d) Run a regression of *Earnings* on *Height*.

(i) What is the estimated slope?

(ii) Use the estimated regression to predict earnings for a worker who is 67 inches tall, for a worker who is 70 inches tall, and for a worker who is 65 inches tall.

(e) Suppose height were measured in centimeters instead of inches. Answer the following questions about the *Earnings* on *Height* (in cm) regression.

(i) What is the estimated slope of the regression?

(ii) What is the estimated intercept?

(iii) What is the R^2?

(iv) What is the standard error of the regression?

(f) Run a regression of *Earnings* on *Height*, using data for female workers only.

(i) What is the estimated slope?

(ii) A randomly selected woman is 1 inch taller than the average woman in the sample. Would you predict her earnings to be higher or lower than the average earnings for women in the sample? By how much?

(g) Repeat (f) for male workers.

[1] As acknowledged by Stock and Watson, these data were provided by Professors Anne Case (Princeton University) and Christina Paxson (Brown University) and were used in their paper "Stature and Status: Height, Ability, and Labor Market Outcomes," *Journal of Political Economy*, 2008, 116(3): 499–532.

(h) Do you think that height is uncorrelated with other factors that cause earnings? That is, do you think that the regression error term, u_i, has a conditional mean of 0 given *Height* (X_i)? Empirical Exercise E5.1 explores one possible explanation for this relationship.

Empirical Exercise E5.1 (adapted from Stock and Watson, pp. 162–163)

Use the data set `Earnings_and_Height` described in Empirical Exercise E4.2 to carry out the following exercises. Part (d) in the original exercise is not assigned here.

(a) Run a regression of *Earnings* on *Height*.

  (i) Is the estimated slope statistically significant?

  (ii) Construct a 95% confidence interval for the slope coefficient.

(b) Repeat (a) for women.

(c) Repeat (a) for men.

(e) One explanation for the effect of height on earnings is that some professions require strength, which is correlated with height. Does the effect of height on earnings disappear when the sample is restricted to occupations in which strength is unlikely to be important?

2 Analytical Exercises

Exercise 5.2 (adapted from Stock and Watson, p. 159)

Suppose a researcher, using wage data on 250 randomly selected male workers and 280 female workers, estimates the OLS regression

$$\widehat{Wage} = \underset{(0.23)}{12.52} + \underset{(0.36)}{2.12} \times Male, \qquad R^2 = 0.06, \quad SER = 4.2,$$

where the numbers in parentheses are standard errors. *Wage* is measured in dollars per hour, and *Male* is a binary variable equal to 1 if the person is male and 0 if the person is female. Define the wage gender gap as the difference in mean earnings between men and women.

(a) What is the estimated gender gap?

(b) Is the estimated gender gap significantly different from 0? Compute the p-value for testing the null hypothesis that there is no gender gap.

(c) Construct a 95% confidence interval for the gender gap.

(d) In the sample, what is the mean wage of women? Of men?

Exercise 5.8 (adapted from Stock and Watson, p. 161)

Suppose (Y_i, X_i) satisfy the least squares assumptions. A sample of size $n = 100$ yields

$$\widehat{Y} = \underset{(10.2)}{43.2} + \underset{(7.4)}{61.5}X, \qquad R^2 = 0.54, \quad SER = 1.52,$$

The numbers in parentheses are standard errors. **Use the large-sample standard-normal approximation**, with the reported standard errors, to construct confidence intervals and conduct hypothesis tests.

(a) Construct a 95% confidence interval for β_0.

(b) Test $H_0 : \beta_1 = 55$ versus $H_1 : \beta_1 \neq 55$ at the 5% level. *Note that the null hypothesis is not zero.* If you use the test-statistic approach for this part: subtract the null value of 55 from the slope estimate and divide by its standard error. Compare the resulting statistic with the two-sided standard-normal critical values at the 5% level.

(c) How do you interpret your test result in part (b) at the 5% significance level? Explain what your decision implies about the evidence concerning $\beta_1 = 55$.

3 True/False Questions

For each statement, determine whether it is true or false and briefly justify your answer.

1. If the data are heteroskedastic, then the regression errors will have different *means* across different values of the regressors.

2. If the data are heteroskedastic, then the *OLS estimate* will differ from the one obtained under the assumption of homoskedasticity.

3. The heteroskedasticity-robust standard error does not estimate the correct standard deviation of the OLS coefficient under homoskedasticity.

4. The confidence interval

$$\left[\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1),\ \hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1)\right]$$

   will still attain 95% coverage of the true parameter β_1 when the sample size is small, say $n = 10$.